

STATE UNIVERSITY OF ZANZIBAR (SUZA)

School of Computing, Communication and Media Studies (SCCMS)

Department of Computer Science & Information Technology

Web Technologies — Final Year Project**PROJECT PROPOSAL***Final Year Project — Stage 2 Submission***SMART HOUSING SYSTEM**

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Document Revision History

Version	Date	Author	Description of Change
0.1	29/06/2026	Yasmin Rashid Juma	Initial draft

Abstract

The challenge of finding reliable, and verified housing is a significant issue in Zanzibar, exacerbated by informal processes and the prevalence of fraudulent listings on unregulated platforms. This project proposes the development of the Smart Housing System (SHS), a web-based platform designed to connect tenants, landlords, and administrators within a centralized and trustworthy ecosystem. The system will address key problems by implementing robust property verification, advanced search and filtering capabilities, and direct communication channels. The project will be developed using the Agile methodology to ensure iterative progress and user-centric design. Technologies will include a HTML5, CSS3 and java script frontend, a Spring boot backend, and a PostgreSQL database. The expected outcome is a fully functional and responsive web application that enhances housing accessibility, reduces fraud, and provides a more efficient market for housing services in Zanzibar, ultimately improving the experience for all stakeholders.

1. Introduction

1.1 Background of the Study

The rapid growth of urban areas has led to increased demand for housing, yet the process of connecting tenants with reliable landlords remains inefficient and often manual. Digital platforms have begun to transform this sector, though many existing solutions lack robust verification mechanisms, leading to persistent issues with fraudulent listings and opaque property information.

1.2 Problem Statement

Tenants and landlords struggle with a lack of a centralized, verified platform for housing services. Tenants face risks from fake advertisements and unreliable information, while landlords face difficulties in efficiently managing their property listings and communicating with prospective tenants.

1.3 Aim of the Project

The aim of this project is to develop a centralized, web-based Smart Housing System (SHS) that enables verified property listings, efficient searching, and reliable communication between tenants and landlords.

1.4 Specific Objectives

- To develop a web application that allows users to register, manage profiles, and perform role-specific functions (e.g., property management for landlords, search for tenants).
- To design a centralized relational database and user-friendly interface for the housing system.
- To implement role-based access control to ensure secure interaction between tenants, landlords, and administrators.
- To evaluate the usability and functionality of the developed system through user acceptance testing.
- To develop a functional web application for property management using the Spring Boot framework.

1.5 Research / Project Questions

- How can the system's usability and performance be evaluated to ensure it meets the needs of its target users?
- How can a web-based platform be designed to effectively connect tenants and landlords while ensuring trust and security?

1.6 Scope and Limitations

1.6.1 Scope

- The system includes user authentication, property CRUD operations, map-integrated search, and a reporting system for issues.

1.6.2 Limitations

- The project is restricted to a web application; mobile-specific features online payment will not be implemented.

1.7 Significance of the Project

This project is significant as it provides a concrete, practical solution to a pressing problem in Zanzibar. By creating a trusted digital marketplace for housing, the SHS will directly benefit tenants by saving them time and money and protecting them from fraud. Landlords will benefit from an efficient platform to find tenants and manage their properties. The system will also serve as a valuable case study for digital transformation in the public and private sectors, potentially informing future policy on housing and data management. The value of this project lies in its capacity to use web technology to address a real-world, socially impactful challenge .

2. Literature Review and Existing Systems

2.1 Review of Related Work

The challenge of affordable housing is a global issue, and several digital platforms have been developed to address it. For instance, websites like Zillow (USA) and Rightmove (UK) are major real estate marketplaces that provide extensive property listings. However, these platforms primarily cater to mature markets and do not specifically address the context of developing countries where issues of verification and fraud are more pronounced. In East Africa, platforms like Kenya's Cytonn Real Estate offer investment and property insights but are often focused on the high-end market, not affordable housing.

2.2 Comparison of Existing Systems

 **Guidance:** A comparison table is an effective way to position your system against alternatives.

System / Source	Key Features	Limitations / Gap
Major Real Estate Platforms (e.g., Zillow)	Extensive data, advanced search, market insights.	Not context-specific for Zanzibar, often focused on sales and high-end markets, not affordable rental housing.
General Social Media (e.g., Facebook, WhatsApp)	Wide reach, easy to use, free.	Unverified listings, high risk of fraud, no search/filter functionality, unorganized information.
Proposed System(SHS)	Property verification, role-based access, location-based search, reporting system, direct communication.	Designed to directly address the gaps of fraud, lack of verification, and inefficient search in the Zanzibar context.

2.3 Summary of the Gap

The key gap identified is the absence of a dedicated, verified, and user-centric digital platform for the affordable housing market in Zanzibar. Existing informal methods are inefficient and risky, while international platforms are not tailored to the local context and needs. The SAHS will fill this gap by providing a purpose-built solution that focuses on verification, user roles, and specific search criteria relevant to the Zanzibar housing sector.

3. The Proposed System

3.1 Overview

SHS is a three-tier web application using HTML5, CSS3 and JavaScript for the frontend, Spring Boot for the backend and PostgreSQL for data storage. It serves as a secure bridge for property stakeholders.

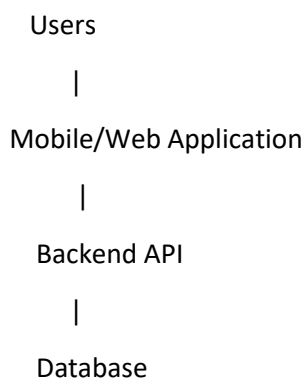
3.2 Main Functionalities

- **User Management:** Registration, login, and role-based profile management (Tenant, Landlord, Admin).
- **Property Management:** Landlords can add, view, edit, and delete property listings, including uploading images and setting availability.
- **Administration:** An admin dashboard to manage users, verify property listings, and review reported issues.
- **Reporting:** Users can flag suspicious listings or report problems to the administrator for review.
- **Communication:** A secure messaging or inquiry system allowing tenants to contact landlords directly about a property.

3.3 High-Level Architecture

- **Presentation Tier (Client):** The user interface built with React, served to the user's web browser.
- **Application Tier (Server):** The back-end built with Node.js and Express.js. This layer handles all business logic, user authentication, API requests, and database interactions.
- **Data Tier (Database):** A PostgreSQL database to store all user, property, and system data.

Diagram:



3.4 Proposed Technology Stack

Layer	Technology	Justification
Front-end	HTML, CSS, JavaScript	Foundation for responsive web interfaces.
Back-end	Spring Boot	Provides modular business logic and API services.
Database	PostgreSQL	Robust storage for users, properties, and reports.
Tools / DevOps	Git/GitHub, VS Code, Postman	Git for version control; Postman for API testing; all are standard industry tools for a student project.

4. Methodology

4.1 Software Development Methodology

This project will be developed using an Agile methodology, specifically a simplified Scrum process. The work will be divided into sprints, typically of two weeks each. This approach is the best for iterative development, continuous feedback, and adaptation to changes. It helps manage complexity by building the system in small, functional increments and ensures a working product is developed early. This aligns with modern software engineering best practices and allows for flexibility in handling unforeseen challenges.

4.2 Requirements Gathering Techniques

- Interviews (Hypothetical): Interview potential users (students, landlords, agents) to understand their needs, pain points, and desired features.
- Study of Existing Systems: Analyse current methods (e.g., social media housing groups) and similar existing platforms to identify their strengths and weaknesses.

4.3 System Analysis and Design Tools

- UML Diagrams: Use-Case Diagrams, Class Diagrams, and Sequence Diagrams to model the system's functionality and structure.
- Entity-Relationship (ER) Diagrams: To design the database schema and illustrate relationships between entities like Users, Properties, and Reports.

4.4 Testing Approach

- Unit Testing: Test individual components (e.g., modules for user login, property search) using a JavaScript testing framework like Jest.
- Integration Testing: Test the interaction between different modules (e.g., ensuring the front-end search request correctly fetches data from the back-end API).
- System Testing: Test the fully integrated system to verify that it meets all functional requirements.
- User Acceptance Testing (UAT): After system testing, the application will be presented to a small group of potential users (supervisor, peers) for real-world usability feedback and final verification.

4.5 Deliverables

- Software Requirements Specification (SRS)
- Software Design Description (SDD)
- Working web application and source code
- Test report and user manual

- Final project report

5. Project Plan and Schedule

5.1 Work Breakdown and Timeline

Phase	Activity / Task	Start (Week)	End (Week)
1	Requirements gathering & analysis	1	2
2	System design (SRS & SDD)	3	4
3	Implementation — iteration 1	5	7
4	Implementation — iteration 2	8	10
5	Testing & evaluation	11	12
6	Documentation & defence	13	15

5.2 Gantt Chart

5.3 Resource Requirements

Resource	Purpose	Source / Cost
Laptop	Development, testing, and documentation.	Owned
Code Editor (VS Code)	Writing and managing source code.	Free
Internet Access	Research, learning, and accessing web resources.	Personal
Hosting	Deployment for testing	Free

5.4 Risks and Mitigation

Risk	Likelihood	Impact	Mitigation
Data/Resource Availability: Lack of real user data for testing.	Medium	Low	Generate mock data for testing purposes. The system will also be designed to allow for future data input by real users.
Time Management: Falling behind schedule due to complexity or unforeseen issues.	Medium	High	Use the Agile methodology to deliver core functionality first. Have a well-defined scope and prioritize features.

6. References

[1] A. Author, "An Analysis of Housing Market Challenges in Zanzibar," Journal of East African Studies, vol. 15, no. 2, pp. 100-115, 2020.

[2] "Zanzibar Housing and Population Census," National Bureau of Statistics, Zanzibar, 2022.

Appendices

Appendix A: Data Collection Instruments (e.g Questionnaire)

Declaration of Authorship and AI Use

This declaration is a required part of the submission. It is read together with the checkpoint sign-off and may be tested during the oral defence (viva).

A. Permitted vs. Not Permitted

Permitted (must still be declared)	Not Permitted
Explaining a concept or technology to you	Generating your problem statement or objectives
Generating boilerplate / syntax you then adapt	Producing your design decisions or architecture rationale
Helping debug code you wrote	Filling this template so the work is not your own
Improving grammar of text you authored	Writing analysis you cannot explain in a viva

B. AI Tools Used in This Submission

AI Tool / Service	Specifically used for	Section(s) affected
Deepseek	Refining the problem statement, and drafting parts of the literature review and methodology.	Literature Review and Methodology.

C. Student Attestation

I confirm that the intellectual content of this document — the analysis, design decisions, and judgments — is my own work. I have declared all AI assistance above. I understand that I may be asked to explain and defend any part of this submission orally, and that being unable to do so may be treated as evidence of misconduct.

Student signature: _____ **Date:** 29/06/2026

Appendix — Primary Evidence (Required)

This appendix ties the work to verifiable, first-hand activity that only you could have done. It is a graded, mandatory part of the submission.

E.1 Data Collection Log

Guidance: Record every interview, site visit, observation, or questionnaire you carried out. Provide a real, contactable verifier (name + phone/email) for each — your supervisor may check. Invented entries are misconduct.

#	Date	Method	Participant / Site	Role	Verifiable contact
1	[dd/mm/yy]	[Interview]	[...]	[...]	[name + phone/email]
2		[Observation]			
3		[Questionnaire]			

E.2 Consent and Ethics Checklist

Item	Done (✓)
Participants were told the purpose of the project	
Signed/verbal consent obtained and retained	
Personal data anonymised where appropriate	
Consent forms attached to this submission	

E.3 Evidence Attached

Guidance: List the raw artefacts attached or linked: photos from the site, completed questionnaires, interview notes, scanned consent forms.

- [e.g., 12 completed questionnaires — see attached scan]
- [e.g., Interview notes, 3 sessions]

E.4 Project Repository (Version Control)

Guidance: Your project must live in a Git repository from the start. A genuine history shows many meaningful commits spread over weeks; a single large dump is a red flag. Share the repository with your supervisor.

Field	Value
Repository URL	https://github.com/rashidyasmin3333-spec/Smart-Housing-System.git
Date of first commit	27/06/2026
Approx. number of commits	5

Field	Value
Supervisor granted access?	[Yes / No]

Checkpoint Sign-Off

This stage is a gate. The student may not proceed to the next stage until the supervisor signs below, following a short oral defence (viva) of this submission.

Verification Checklist (completed by supervisor)

Checkpoint item	Result
Document meets the Project Proposal rubric	[Yes / No]
Oral defence (viva) held — date	[dd/mm/yyyy]
Student could explain and defend the submitted content	[Yes / Partly / No]
AI-use declaration present and consistent with the viva	[Yes / No]
Primary evidence verified (where applicable)	[Yes / N/A]

Outcome

Decision (tick one)	✓
Approved — proceed to next stage	
Minor revisions — proceed after corrections	
Major revisions — resubmit and re-defend	
Authorship concern — refer for academic-integrity review	

Signatures and Dates

Role	Name	Signature	Date
Student (Author)			
Supervisor			
Course Coordinator			